

Subject: → Need of IP address over mac address? //

Host

IANA
(Internet Assigned Network Authority)

IP

→ IP v4 → 32 bit packets
→ IP v6

provides IP

divides IP in 5 classes

Using for unicast
use for multicasting

←	Class A	no. of ip's required
	— B	
	— C	
←	— D	
	— E	

for defence, private ip's

→ 10.08.11.10
(32-bit)

Repⁿ → 1. Binary
00001010
2. Dotted

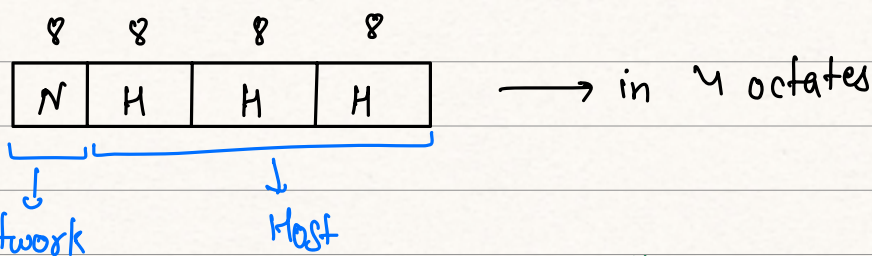
Subject: / /

IANA $\rightarrow$ gives the hierarchy approach. A computer on the internet is addressed using the tuples / types:-

1. Network no. (Assigned and managed by central authority)

2. Host no. (Assigned and managed by local network authority)

* When routing a packet to the destination network only the network number is looked at



$\hookrightarrow 24 \rightarrow 2^{24} - 2$

$\downarrow$ (DBA + NetID)
IP

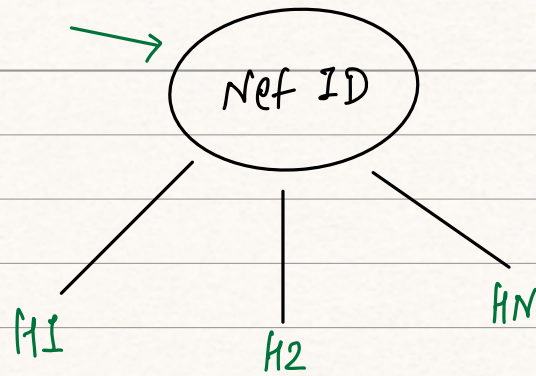
Among 8, a bit is reserved and rest 7 are network

1 + 7
 $\downarrow$ reserved $\downarrow$ N

DBA $\rightarrow$ Direct Broadcasting address. $\rightarrow$ broadcast a message in network (IP)

LBA $\rightarrow$ Local Broadcasting address. (last IP)

Subject:



Net ID (IP)
→ 1st IP
↳ particularly send a message to one.

→ Special case in Class A



*

DHCP

reserved → 1 + 7
↓
2⁷ - 2
↳ two reserved here

1. one for those system which don't have ip
2. one for DHCP (last)

Class A range :-

starts → 00000000 : H . H . H

ends → 0 | 1 | 1 | 1 | 1 | 1 | 1 : H . H . H
 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰

0 — 127

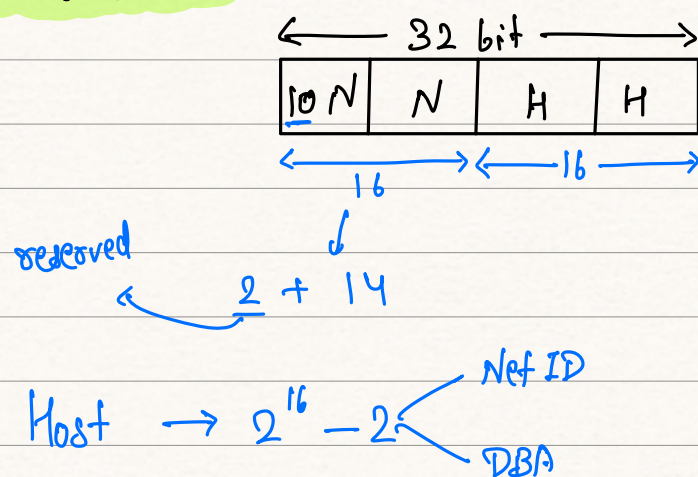
so,

0.0.0.0 — 127.255.255.255

in special case
this reserved for the system that don't have ip address
so range updates from
1 — 127

Subject:

Class B :-



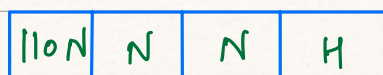
Network $\rightarrow 2^{14}$

$\hookrightarrow 100000000 . 000000000$

↓ range

11111111 .

Class - C :-



$110(2^{21})$

$\hookrightarrow (2^8 - 2)$

* Host bits = $2^8 - 2$

* Network bits = 2^{21}

range:- 110 00000 (192)

↓
110 11111 (223)

192.0.0.0 — 223.255.255.255

Subject:

Class D

Class E

/ /

1110	
------	--

1111	
------	--

224 - 239

A → 1 - 126

B → 128 - 191

C → 192 - 223

D →

E →

* Network mask: Network mask is a mathematical tool which is used for solving networking problem.

By default network mask:-

Class A : 255.0.0.0

B : 255.255.0.0

C : 255.255.255.0

* Bitwise AND b/w I/P & mask will give Net ID

Q. $IP_1 = 201.55.79.89$, Find Net ID



Class C & AND with 255.255.255.0

201.55.79.89

255.255.255.0
—

Subject:

89 → 01011001

79 → 01001111 / /

0 → 00000000

255 → 11111111

00000000

01001111

55 → _____

255 → 11111111

201 → _____

255 → 11111111
